

The prompt

Based on the current running result of the code (Please refer to the details of dataset from <https://www.gapminder.org/data/documentation/>, create an executive summary following the instructions:

1. Keep the summary one-page-long (~ 500 words)
2. Explain (or substitute with everyday concept) the statistic terms in order to face toward general audience
3. Have a good combination of numbers and explanations, and make distinction between what's important and what's not
4. Use formal business language

The response

Executive Summary: Country-Level Daily Income in 2010

An analysis of Gapminder data for 132 countries examines how average daily income per person in 2010 relates to human development, household energy use, mobile phone subscriptions, maternal mortality, and whether a country has a coastline. The model accounts for approximately 84% of the differences in *relative income levels* among the countries analyzed. That is a strong fit to the data used to build it, but it does not establish that these factors cause income to rise or fall.

The clearest relationship is with the Human Development Index, a score that summarizes health, education, and living standards. In the model, a country scoring 0.1 point higher on this index has approximately 63% higher daily income when the other measured characteristics are held constant. **This result requires particular caution:** income is itself part of the index. The model therefore partly uses an income-related measure to account for income. The strength of this relationship should not be presented as evidence that improvements in health or education alone would produce a comparable income gain. ([Human Development Reports](#))

Geography and energy use show smaller, but distinct, relationships. After accounting for the other measured characteristics, landlocked countries have approximately 23% lower daily income than countries with a coastline. Countries where residential use represents a larger share of energy use also tend to have lower income. Roughly doubling that energy-use measure corresponds to about 11% lower income in the model. These comparisons describe patterns across countries in one year. They do not show that obtaining coastal access or changing household energy consumption would, by itself, change income by those amounts.

Two measures offer less evidence of a separate relationship with income once the other characteristics are considered. Countries with more mobile phone subscriptions per 100 people tend to be wealthier in broad comparisons, but the model does not establish a clear additional association for subscriptions. The same is true of maternal mortality, the number of maternal deaths relative to live births. These findings should not be read as evidence that connectivity or

maternal health is unimportant. Rather, the available data do not clearly distinguish their separate relationships with income from the relationships captured by the other measures.

The scope of the analysis also matters. Although the daily-income file contains 193 countries, only 132 have a complete set of the measures required for this model. Countries omitted because an indicator is unavailable may differ systematically from those included. Each country also contributes one observation regardless of population size, so the results describe country-level comparisons, not the experience of the average person worldwide. The analysis uses 2010 data only and cannot show how these relationships changed over time.

Finally, the reported 84% fit describes how closely the model matches the countries used to estimate it; its accuracy on new data has not been tested here. Gapminder notes that some of its series combine sources and fill gaps, with uncertainty that can be substantial. The appropriate business conclusion is therefore measured: the analysis identifies a strong *descriptive* income pattern and a notable difference associated with landlocked geography, while the human development result is partly built into the way that index is defined. Decisions about interventions or forecasts would require additional analysis and independent validation.

a. Did the LLM or not do well at describing the variables used in the project and what questions you were trying to answer? Why?

Yes, in the first sentence it is able to identify the variables – “human development, household energy use, mobile phone subscriptions, maternal mortality, and whether a country has a coastline.” It puts the actual variables we are using in context, as seen above. This is an important part of the summary so it did a good job there. For the question itself all it does is say we are looking at how the variables and the daily income are related. It could be more descriptive here.

b. Did the LLM do well or not at describing the methods used in the project? Why?

The LLM didn't describe the methods used in this project. This is probably due to the fact that we never asked it to. It goes right into the correlations between the respective variables. It never talks about our process of importing the data, cleaning it, creating a log of the valuables, and making a regression line using lm.

c. Did the LLM do well or not at picking the most important results of your project? Why?

It didn't really pick out any results for our project. Rather, it gave a brief summary of each variable and its correlation to daily income. I think that this approach worked well to give a summary of our project results.

d. Did the LLM do well or not at making the result in the style of an executive summary and an appropriate audience? Why?

Yes it did. It has broken up into paragraphs that are organized and easy to understand. It flows well as it breaks down the different parts of the summary. The explanations build so that it is not just jumping around different parts of the project. It did well to keep the word count around 500 and used professional and descriptive language. This is perfect for an audience that is professional but not necessarily technical. m